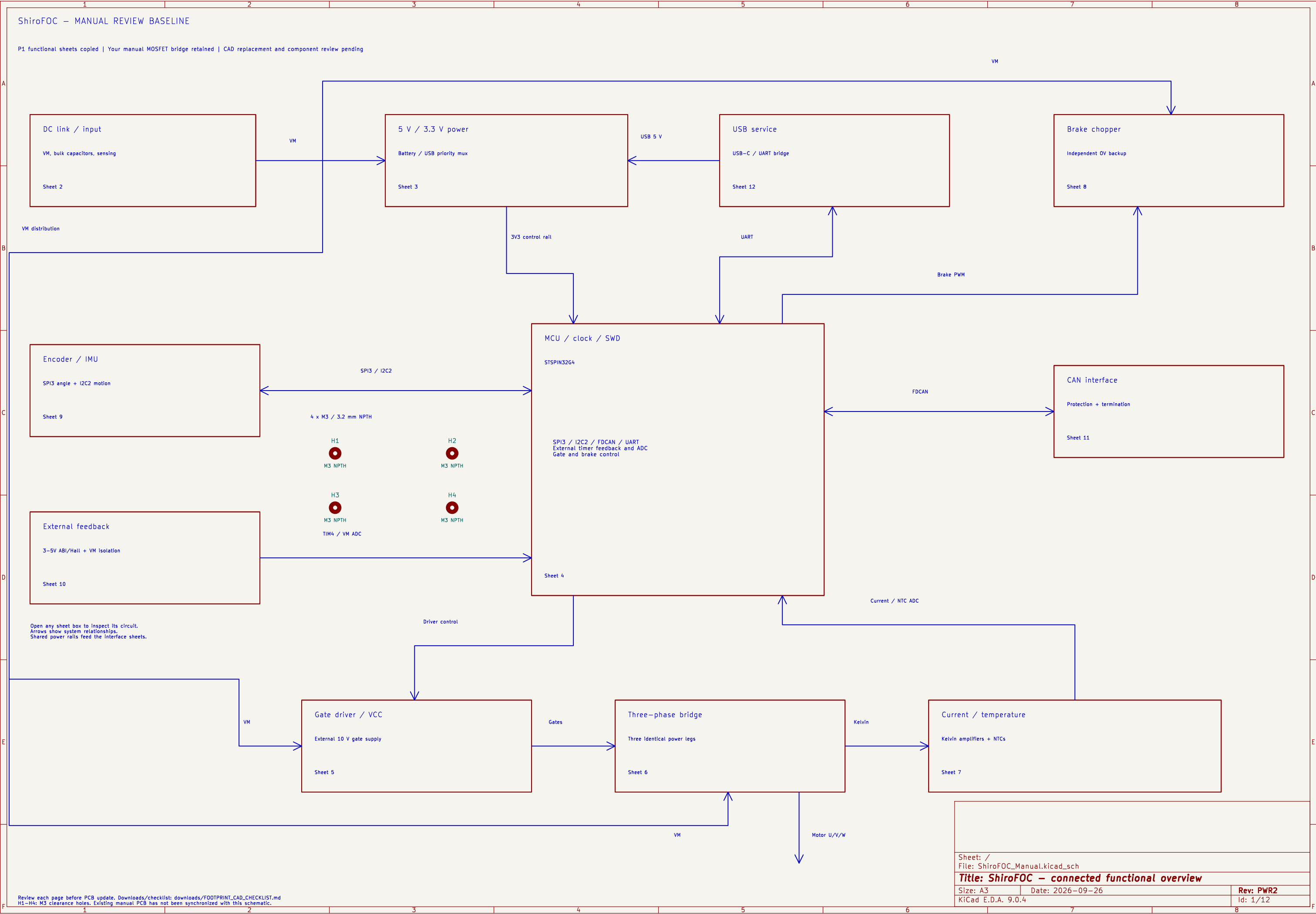
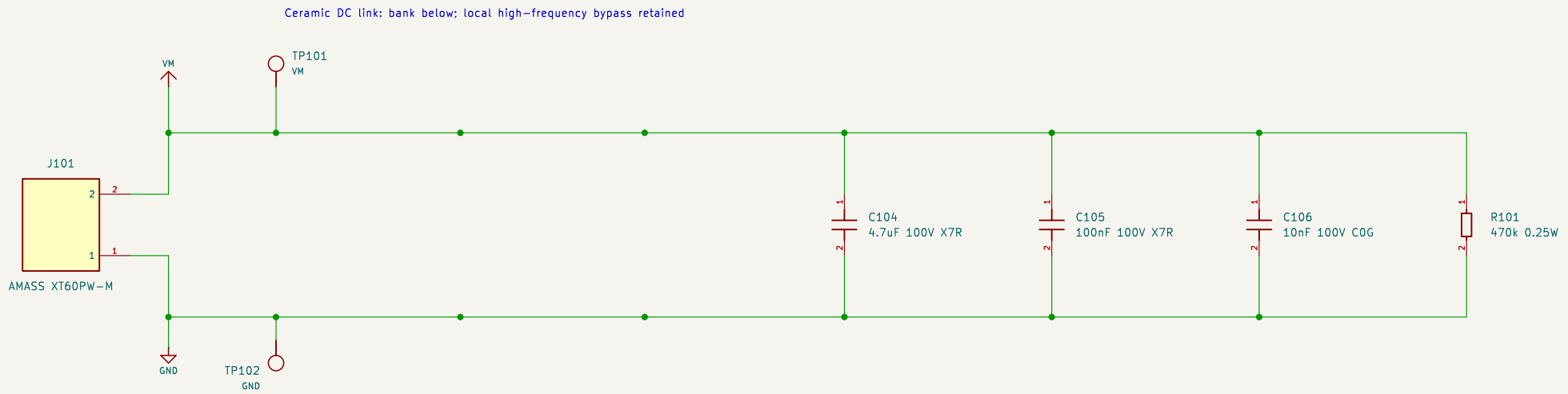


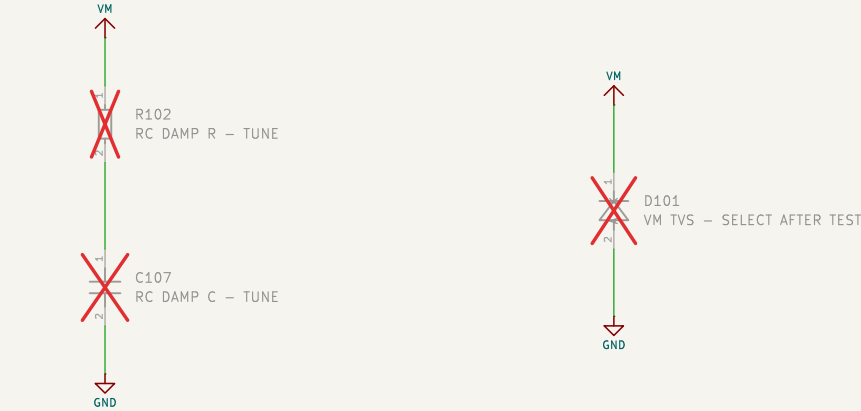


BATTERY / DC LINK

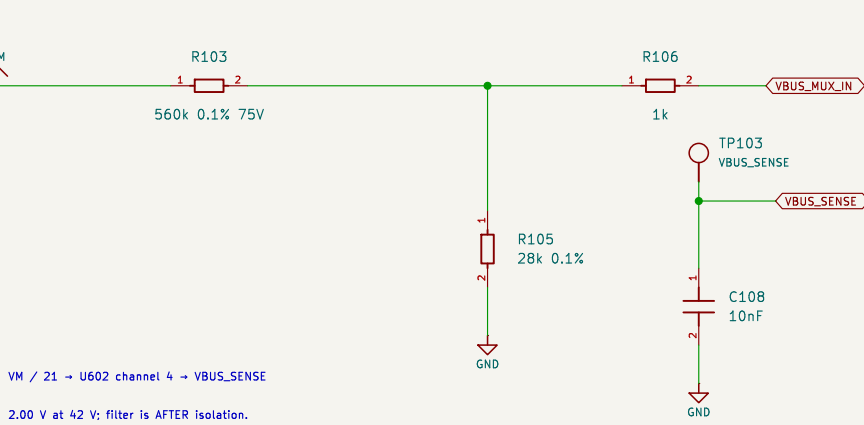
18–42 V operating bus • keyed XT60 • external fuse / precharge required



OPTIONAL INPUT DAMPING

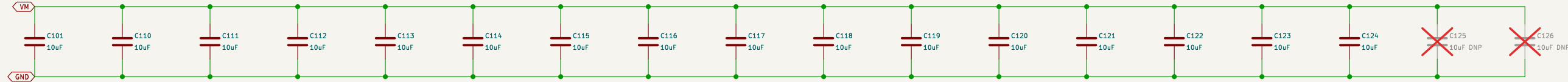


BUS DIVIDER / ISOLATED ADC FILTER

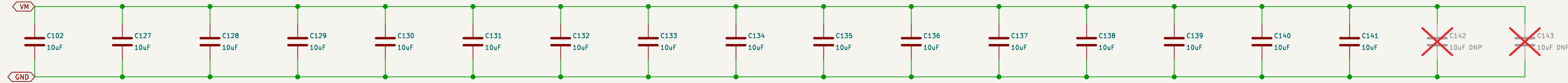


DNP: choose clamp and damping only after cable / hot–plug tests.

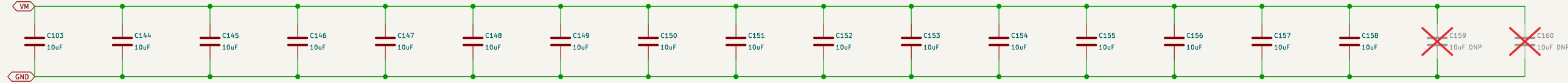
BANK U: 16 FIT + 2 DNP | 10 uF / 100 V / X7S / 1210 | CL32Y106KCVZNWE



BANK V: 16 FIT + 2 DNP | 10 uF / 100 V / X7S / 1210 | CL32Y106KCVZNWE



BANK W: 16 FIT + 2 DNP | 10 uF / 100 V / X7S / 1210 | CL32Y106KCVZNWE

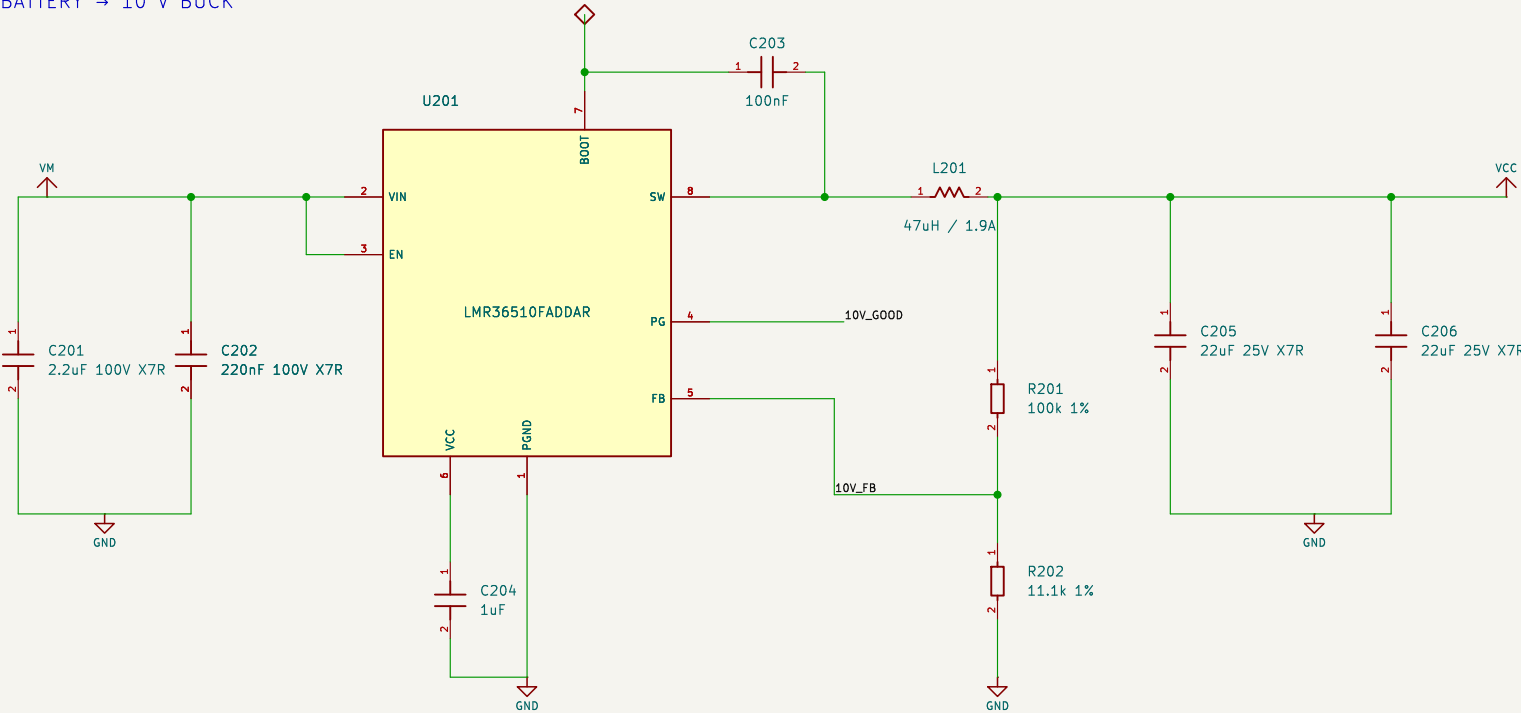


48 fitted = 480 uF nominal; approximately 160 uF typical at 42 V before tolerances.
Six optional positions add 60 uF nominal. This bank replaces C101–C103 electrolytics.
Validate cable resonance, regeneration, ripple heating and transient margin on hardware.

48 fitted MLCs; six optional positions; battery damping retained		
ShiroFOC		
Sheet: /DC link / input/ File: 01_DC_Link.kicad_sch		
Title: Battery input, DC link and bus measurement		
Size: A3	Date: 2026–10–04	Rev: A2–Ceramic
KiCad E.D.A. 9.0.4		Id: 2/12

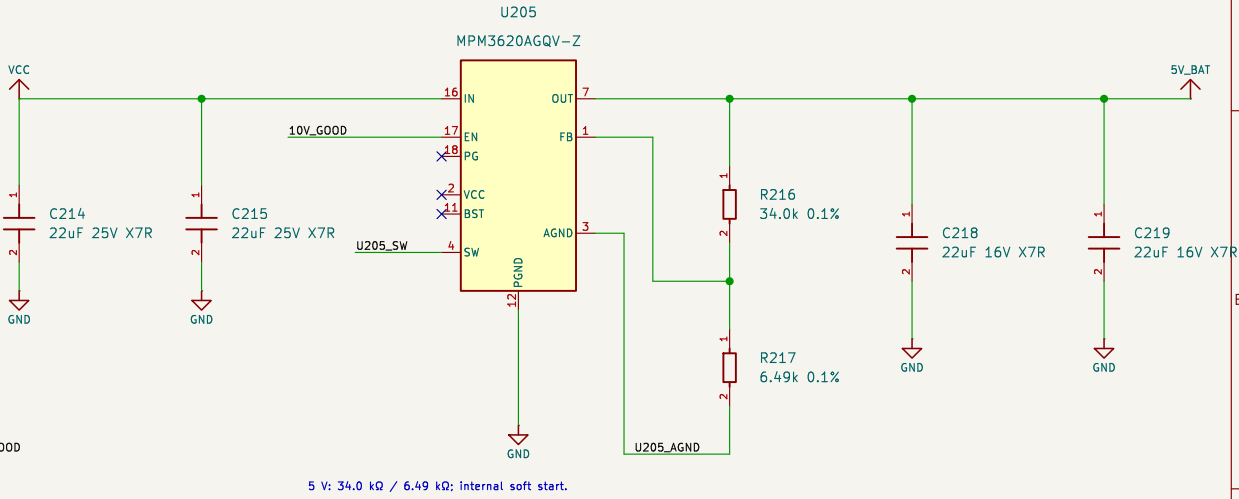
AUXILIARY POWER – USB HAS PRIORITY

BATTERY → 10 V BUCK



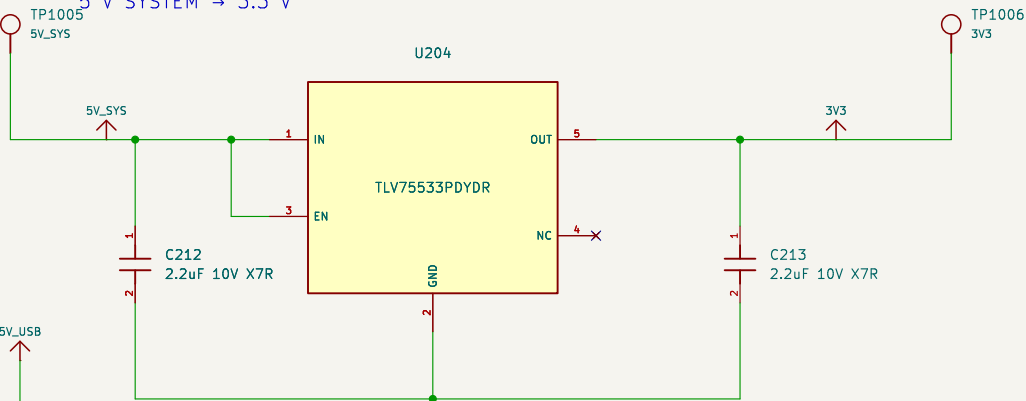
10 V → 5 V / INTEGRATED INDUCTOR

U205 VCC is its internal 4.9 V output; leave open.
AGND serves only the FB divider; joined to PGND inside U205.



5 V: 34.0 kΩ / 6.49 kΩ; internal soft start.

5 V SYSTEM → 3.3 V

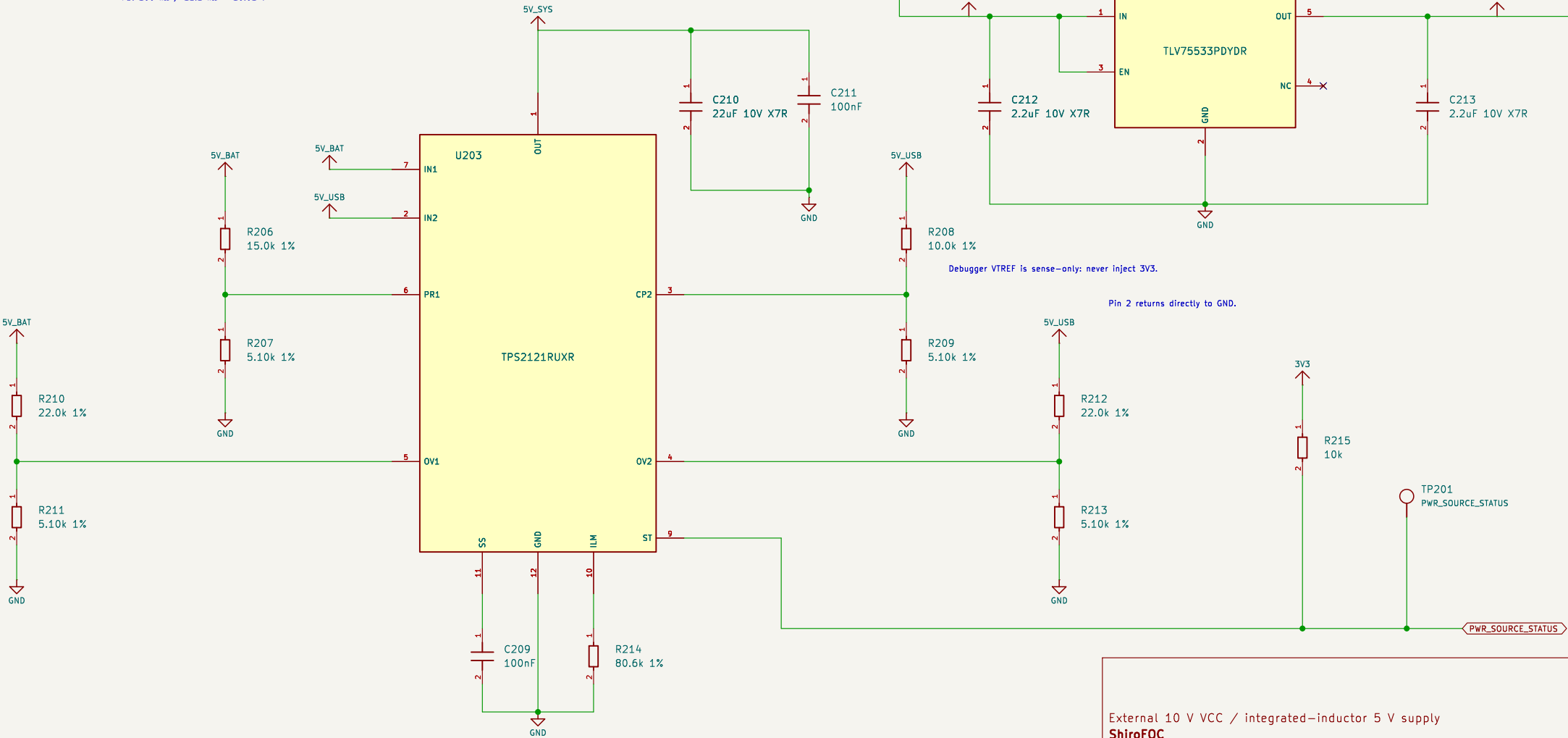


Debugger VTREF is sense-only; never inject 3V3.

Pin 2 returns directly to GND.

USB-PRIORITY POWER MUX

5V_USB arrives from the USB service sheet.



At 5 V: PR1 = 1.269 V; CP2 = 1.689 V.

USB wins the priority comparison. Coarse OV = 5.63 V typical.

80.6 kΩ sets approximately 1.5 A current limit.

VCC is the external 10 V gate supply. Local caps remain at the driver.

U201 PG enables U205 through 100 kΩ / 100 kΩ; EN stays below 6 V.

External 10 V VCC / integrated-inductor 5 V supply

ShiroFOC

Sheet: /5 V / 3.3 V power/

File: 02_Auxiliary_Power.kicad_sch

Title: 10 V / 5 V / 3.3 V auxiliary supplies

Size: A3

Date: 2026-09-26

Rev: PWR3

KiCad E.D.A. 9.0.4

Id: 3/12

CONTROLLER / CLOCK / DEBUG

A

B

C

D

E

F

A

B

C

D

E

F

CAN

UART

SPI3 / I2C2

FEEDBACK

IMU IRQ

BRAKE

MCU / ANALOG SUPPLIES

U4 units: A digital, B supplies, C gate driver, D/E/F current amplifiers. Same STSPIN32G4 package.

U4A

STSPIN32G4

U4B

STSPIN32G4

Y301

24MHz 7pF

24 MHz / 7 pF load
10 pF COG starting caps
Case pads 2 and 4 = GND

D301

RED

STATUS LED: PC15, active low / open drain.
R304 limits LED sink current; keep LSE disabled.
PC2 is now ADC12_IN8 for the external brake NTC.

TC2030-NL: 6 pogo contacts, 3 locating holes.
VREF is sense-only; power board separately.
Pin 6 (SWO) is NC; PB3 is encoder SPI3 SCK.
Keep probe approach and locating holes clear.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /MCU / clock / SWD/
File: MCUclockSWD.kicad_sch

Title: Controller, clock, reset and debug

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

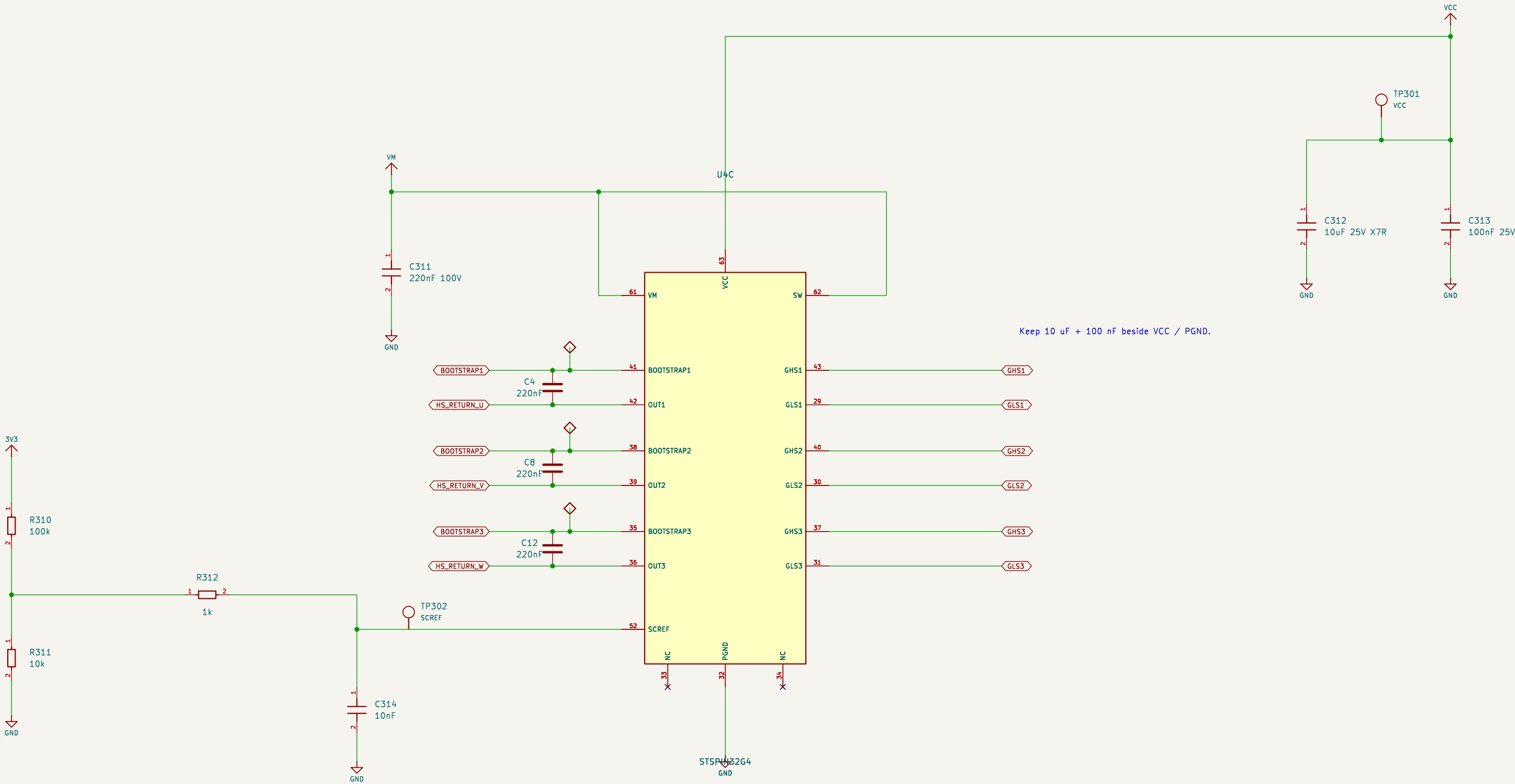
KiCad E.D.A. 9.0.4

Id: 4/12

GATE DRIVER / EXTERNAL 10 V SUPPLY

INTERNAL BUCK BYPASSED: SW → VM

VDS FAULT REFERENCE



SCREF is a VDS fault threshold (~0.30 V), not a precision phase–current limit.
Validate coarse COMP1/2/4 trip and TIM1 break before enabling PWM.

External VCC = 10 V from the auxiliary sheet. SW tied to VM per ST Figure 7.
REGIN and REG3V3/VDD use external 3V3 (unit B). USB alone does not power the gates.

C4/C8/C12: place at BOOTx / OUTx pins. HS_RETURN_U/V/W go directly to MOSFET SH, paired with gate traces.

External 10 V VCC / integrated–inductor 5 V supply

ShiroFOC

Sheet: /Gate driver / VCC/

File: 04_Gate_Driver.kicad_sch

Title: STSPIN gate driver / external 10 V

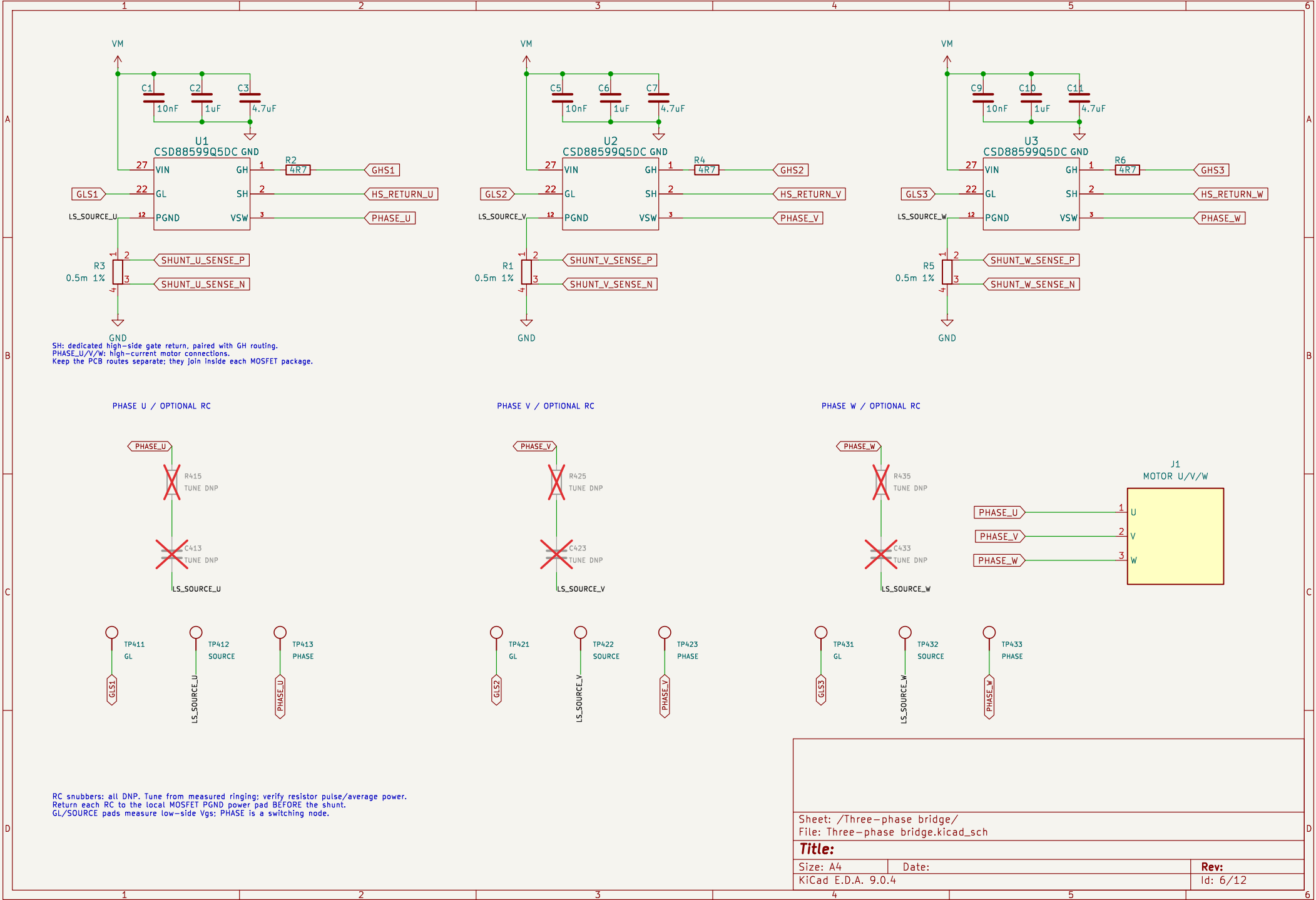
Size: A3

Date: 2026–09–26

Rev: PWR2

KiCad E.D.A. 9.0.4

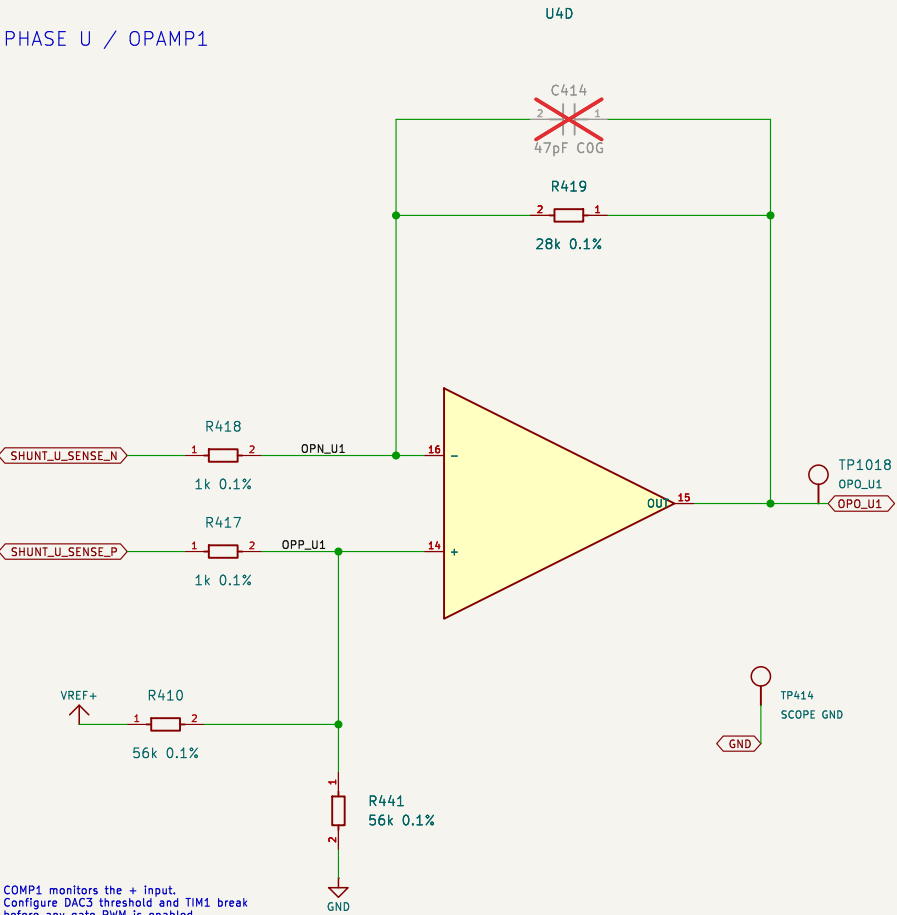
Id: 5/12



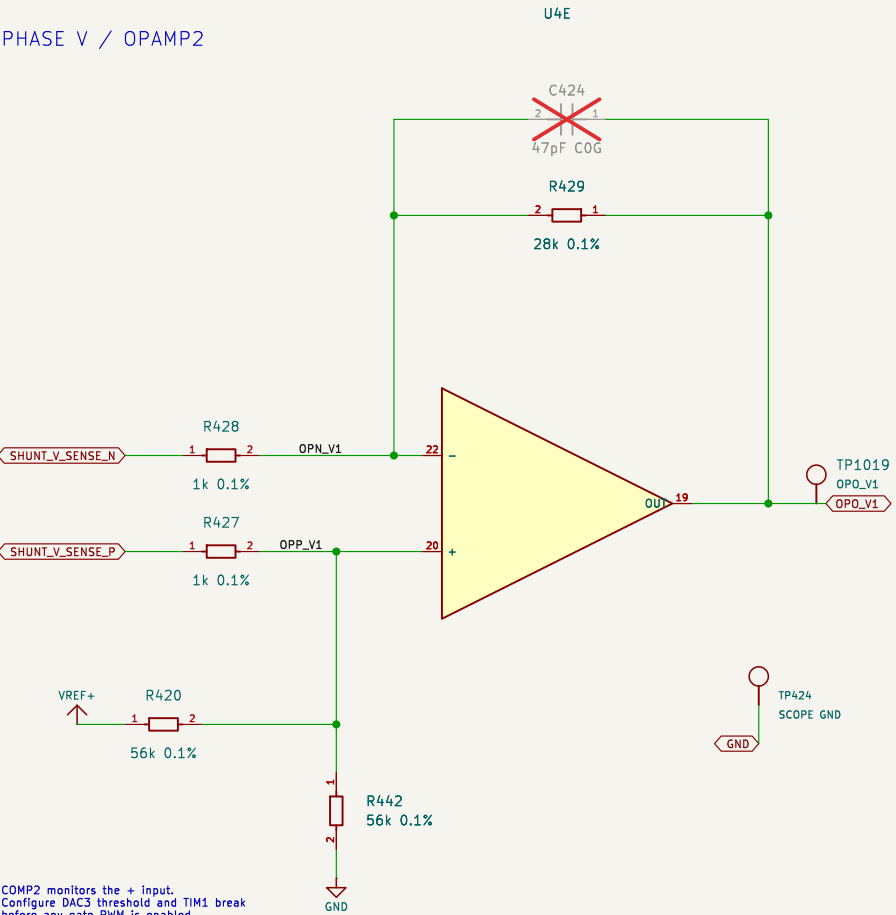
MATCHED CURRENT CHANNELS AND LOCAL MOSFET TEMPERATURE

$V_{OUT} = V_{REF}/2 + 28 \times (SENSE+ - SENSE-)$. At 0.5 mOhm: 14 mV/A. Bias resistors establish the bipolar zero-current point.

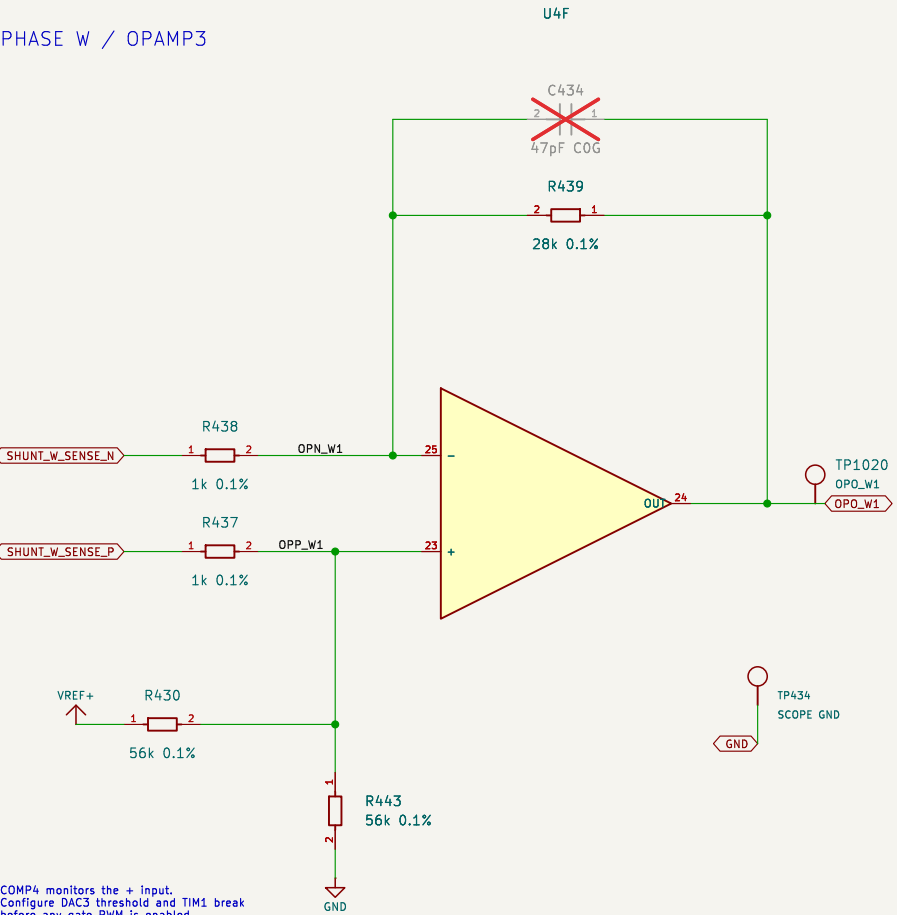
PHASE U / OPAMP1



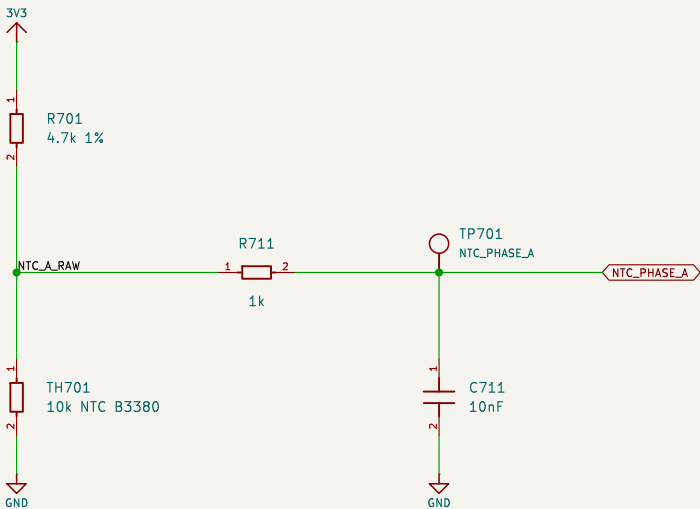
PHASE V / OPAMP2



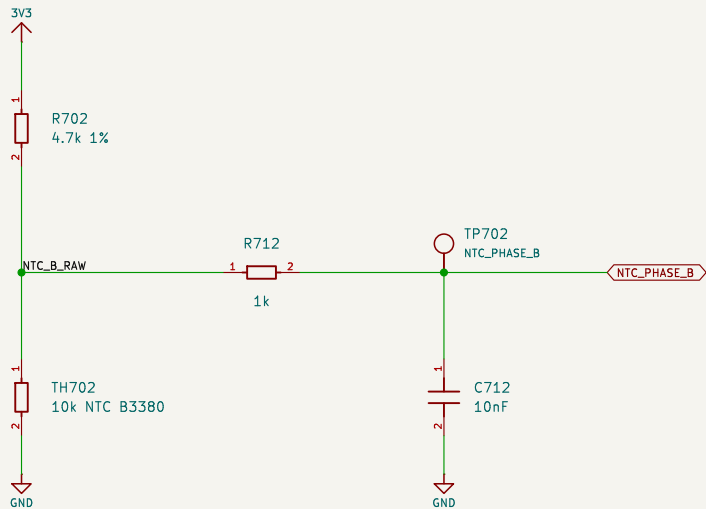
PHASE W / OPAMP3



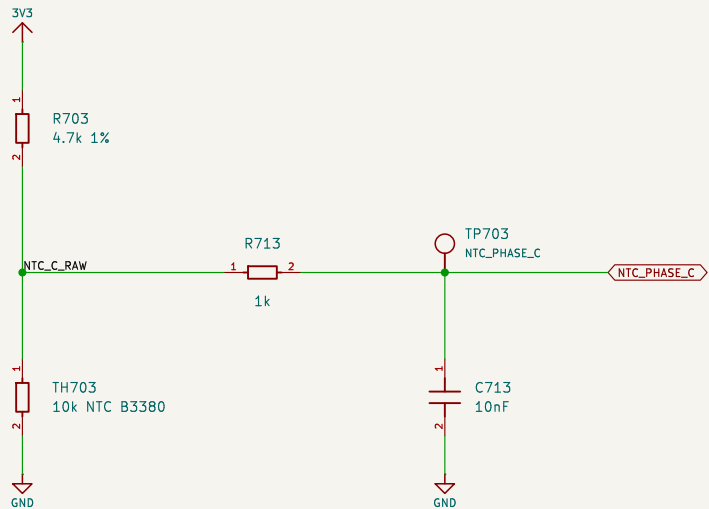
NTC A – place beside phase U



NTC B – place beside phase V



NTC C – place beside phase W



NTC faults: open -> ADC near 3V3; short -> ADC near GND. Use the hottest valid channel and calibrate protection against measured temperature.

R1 / R3 / R5: LR2512D-3W-0.5mR-1%, 0.5 mOhm. Target 25 A RMS with cooling; validate thermally. Scale: 14 mV/A.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Current / temperature/

File: 06_Current_Temperature.kicad_sch

Title: Three Kelvin current amplifiers / phase temperature

Size: A3

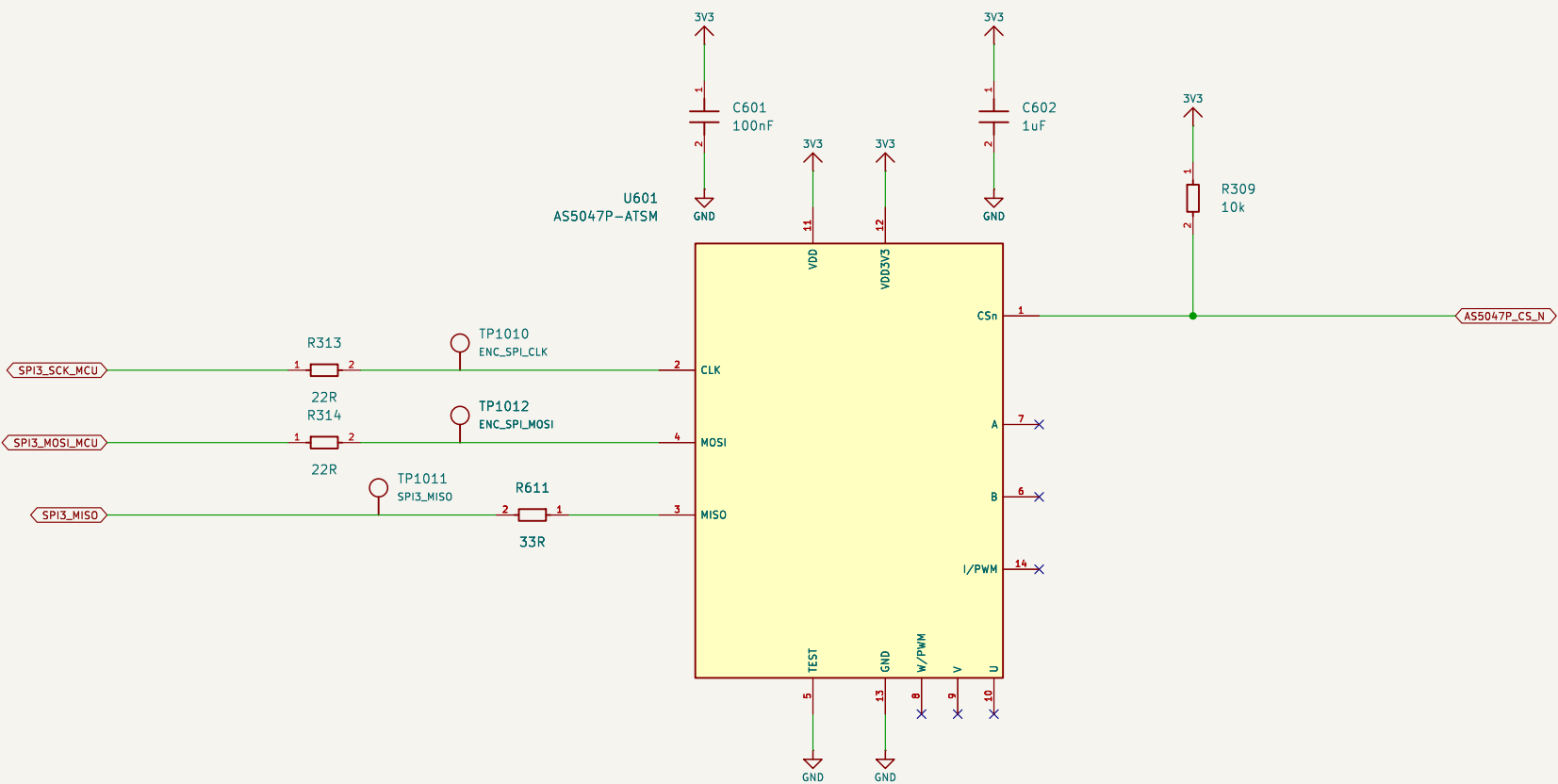
Date: 2026-09-16

Rev: A1-Drawing

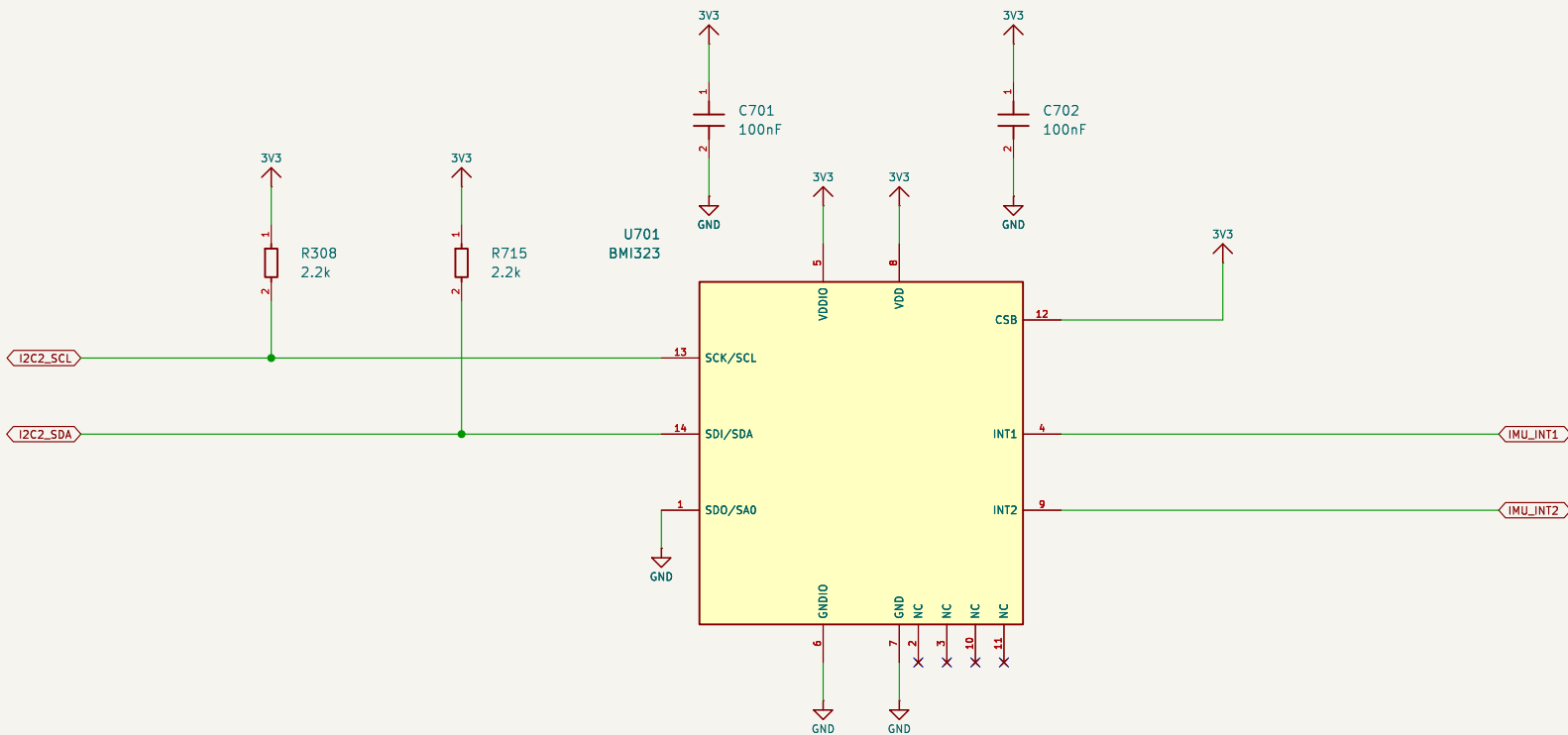
KiCad E.D.A. 9.0.4

Id: 7/12

POSITION + MOTION: dedicated encoder SPI3; independent IMU I2C2



AS5047P ABSOLUTE ANGLE
SPI mode 1: start at about 5 MHz.
Read angle + check parity/error flags.
No ABI count tracking for this sensor.
Align package center with shaft magnet.



BMI323: I2C ONLY.
CSB tied high; SDO low -> address 0x68.
Configuration and samples both use I2C.
400 kHz; target 1 ms sample period.
Reads include two dummy bytes.
Mark sensor axes on the PCB.

R313/R314 near MCU; R611 near encoder MISO. Retained local supply bypass.
I2C pull-ups: 2.2k to 3V3. Target total bus capacitance <=150 pF at 400 kHz.
Keep sensor wiring away from phase nodes and gate-drive loops.

Dedicated sensor buses; interface revision

ShiroFOC

Sheet: /Encoder / IMU/

File: 08_SPI_Sensors.kicad_sch

Title: Absolute encoder SPI / IMU I2C

Size: A3

Date: 2026-09-26

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

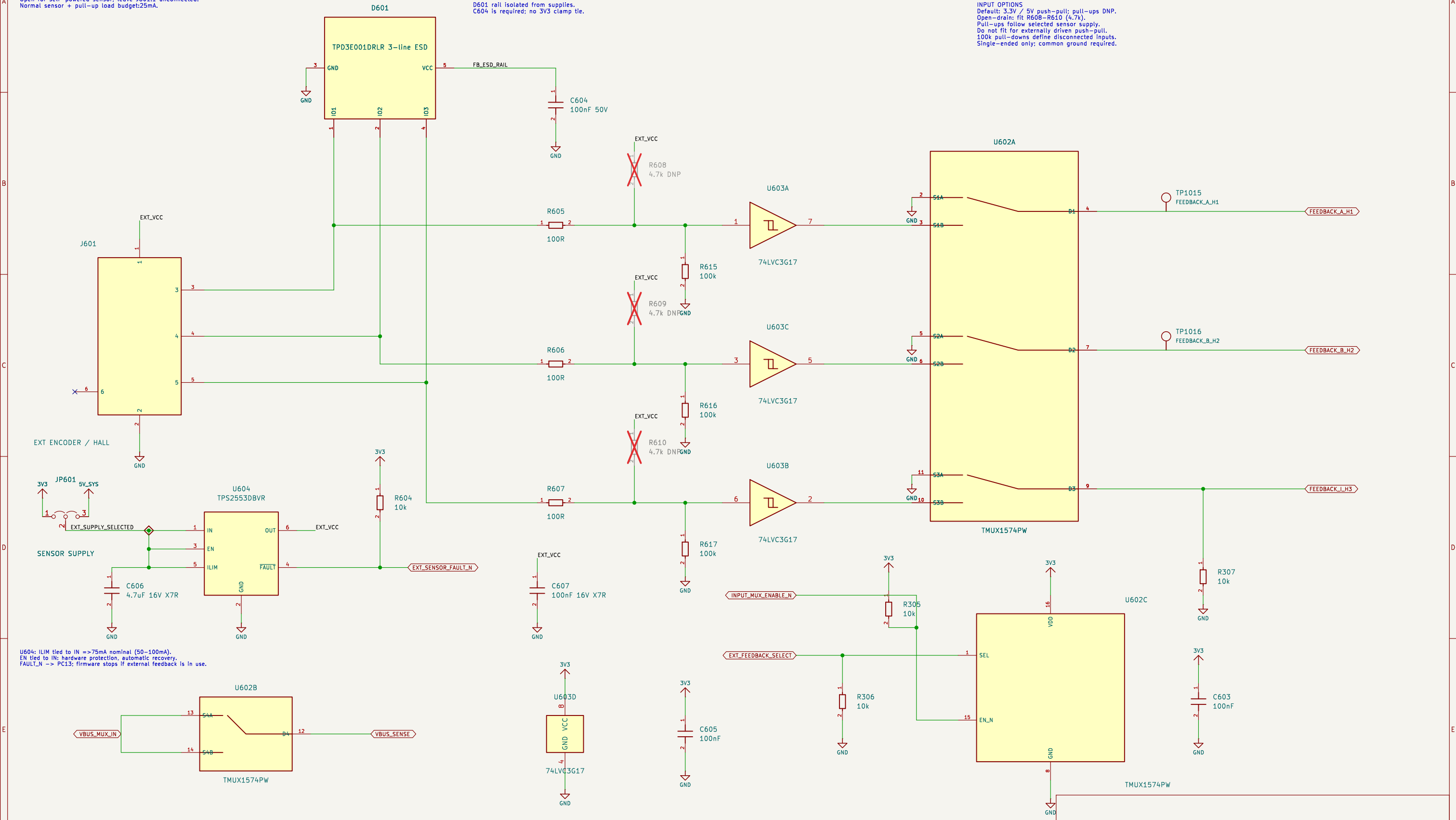
Id: 9/12

EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

External sensor power is protected by U604 below.
JP601: solder ONE pair:1-2=3V3,2-3=5V.
Open for self-powered sensor; leave J601.1 unconnected.
Normal sensor + pull-up load budget:25mA.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

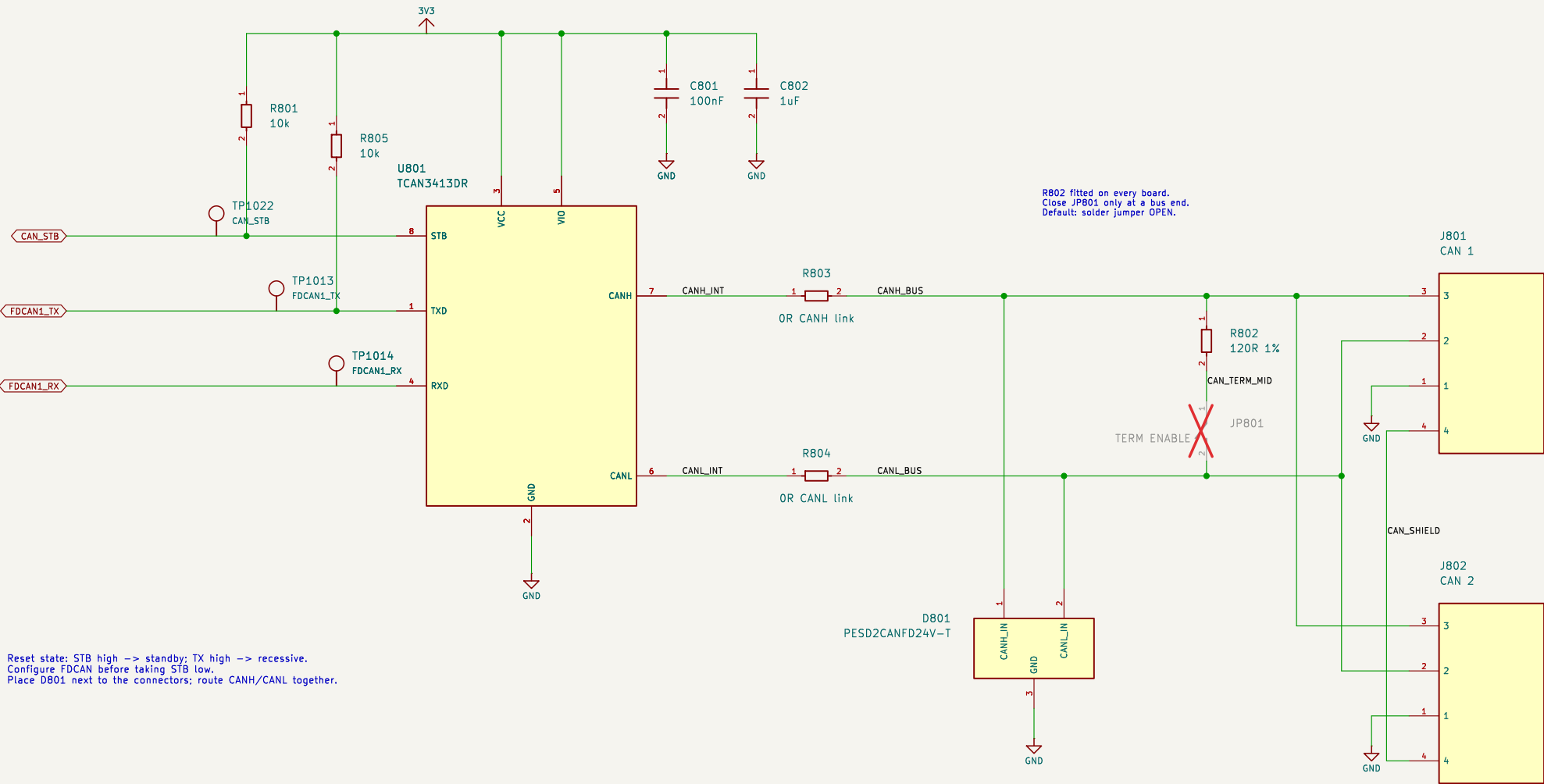
INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k).
Pull-ups follow selected sensor supply.
Do not fit for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.



Reset: mux disabled, PBR/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision		
ShiroFOC		
Sheet: /External feedback/ File: 09_Feedback_Selection.kicad_sch		
Title: External Hall / ABI conditioning and reset isolation		
Size: A3	Date: 2026-09-26	Rev: A1-Drawing
KiCad E.D.A. 9.0.4		Id: 10/12

CAN: controller -> transceiver -> protected daisy-chain bus



Reset state: STB high -> standby; TX high -> recessive.
Configure FDCAN before taking STB low.
Place D801 next to the connectors; route CANH/CANL together.

Both headers: 1 GND | 2 CANL | 3 CANH | 4 shield/drain. Pin 4 carries no supply.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /CAN interface/

File: 10_CAN.kicad_sch

Title: CAN / CAN-FD interface

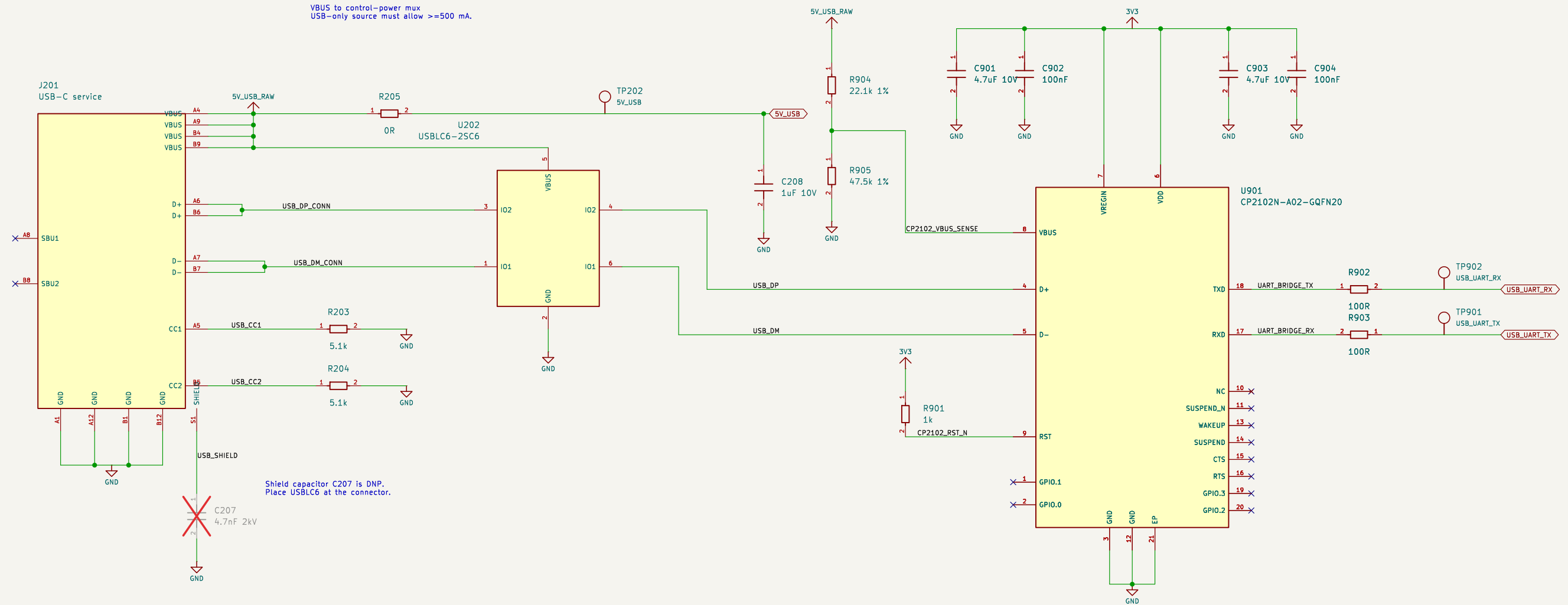
Size: A4 Date: 2026-09-16

KiCad E.D.A. 9.0.4

Rev: A1-Drawing

Id: 11/12

USB service: connector, protection and bridge in one circuit



UART labels are named from the MCU point of view.
TXD from U901 drives MCU RX. RTS/CTS are unused.
SWD recovery and NRST are on the MCU sheet; BOOT0 is option-byte controlled.

LAYOUT: route the USB pair continuously from J201 through U202 to U901; avoid stubs.
C901/C902 belong at VREGIN; C903/C904 belong at VDD. Both pin groups use the same 3V3 rail.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /USB service/

Title: USB-C service / UART bridge

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 12/12